

Name - Saikat Sheet
Batch - 25SUB5936_Linux System Programming
ID - 68500

Task 9:

Directory Operations in C

- Develop a C program to perform directory operations:
- Use system calls to list all files and directories in the current working directory.
- Change the current working directory to "/tmp".
- Create a new directory named "testdir" within "/tmp" and set its permissions to 755.

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <dirent.h> // For directory stream operations
#include <unistd.h> // For chdir()
#include <sys/stat.h> // For mkdir()
#include <errno.h> // For error handling

int main() {
    struct dirent *entry;
    DIR *dp;

    // List all files and directories in the current working directory
    printf("Listing contents of current directory:\n");
    dp = opendir(".");
    if (dp == NULL) {
        perror("opendir");
        return 1;
    }
}
```

```

while ((entry = readdir(dp)) != NULL) {
    printf(" %s\n", entry->d_name);
}
closedir(dp);

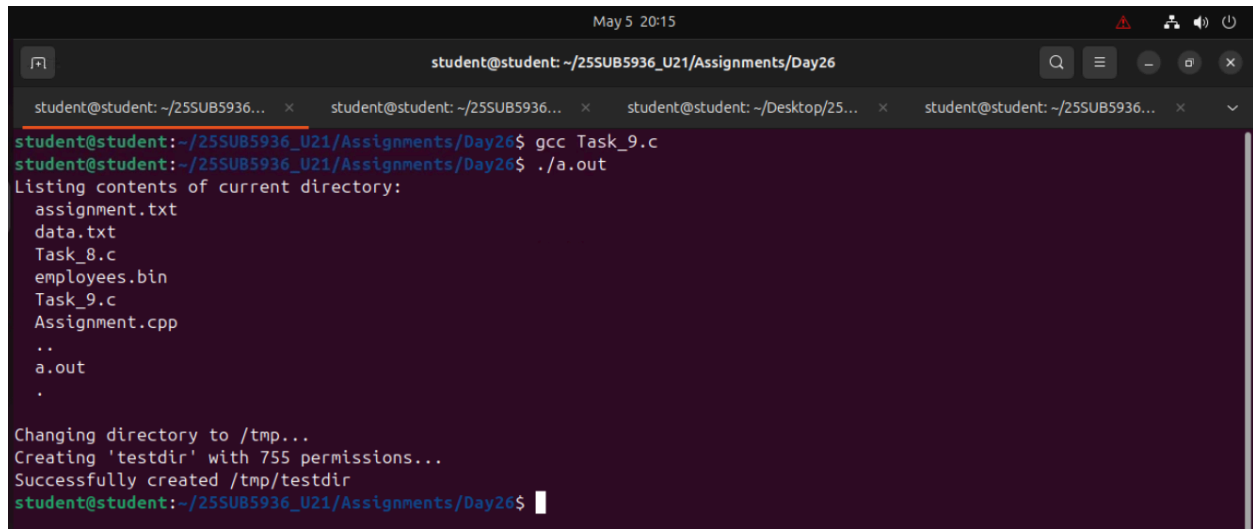
// 2. Change the current working directory to "/tmp"
printf("\nChanging directory to /tmp...\n");
if (chdir("/tmp") != 0) {
    perror("chdir");
    return 1;
}

// 3. Create a new directory named "testdir" with permissions 755
// 0755: Owner (Read/Write/Execute), Group (Read/Execute), Others
(Read/Execute)
printf("Creating 'testdir' with 755 permissions...\n");
if (mkdir("testdir", 0755) != 0) {
    if (errno == EEXIST) {
        printf("Note: 'testdir' already exists.\n");
    } else {
        perror("mkdir");
        return 1;
    }
} else {
    printf("Successfully created /tmp/testdir\n");
}

return 0;
}

```

Output:



```
May 5 20:15
student@student: ~/25SUB5936_U21/Assignments/Day26
student@student: ~/25SUB5936_U21/Assignments/Day26$ gcc Task_9.c
student@student: ~/25SUB5936_U21/Assignments/Day26$ ./a.out
Listing contents of current directory:
assignment.txt
data.txt
Task_8.c
employees.bin
Task_9.c
Assignment.cpp
..
a.out
.

Changing directory to /tmp...
Creating 'testdir' with 755 permissions...
Successfully created /tmp/testdir
student@student: ~/25SUB5936_U21/Assignments/Day26$
```